

Bayesian Linear Models

Sudipto Banerjee

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Department of Biostatistics, Fielding School of Public Health, University of California, Los Angeles

Linear Regression

- Linear regression is, perhaps, *the* most widely used statistical modeling tool.
- It addresses the following question: How does a quantity of primary interest, y , vary as (depend upon) another quantity, or set of quantities, x ?
- The quantity y is called the *response* or *outcome variable*. Some people simply refer to it as the *dependent variable*.
- The variable(s) x are called *explanatory variables*, *covariates* or simply *independent variables*.
- In general, we are interested in the conditional distribution of y , given x , parametrized as $p(y \mid \theta, x)$.

- Typically, we have a set of *units* or *experimental subjects* $i = 1, 2, \dots, n$.
- For each of these units we have measured an outcome y_i and a set of explanatory variables $x_i^\top = (1, x_{i1}, x_{i2}, \dots, x_{ip})$.
- The first element of $x_i^\top$ is often taken as 1 to signify the presence of an “intercept”.
- We collect the outcome and explanatory variables into an $n \times 1$ vector and an $n \times (p + 1)$ matrix:

$$y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}; \quad X = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1p} \\ 1 & x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{np} \end{bmatrix} = \begin{pmatrix} x_1^\top \\ x_2^\top \\ \vdots \\ x_n^\top \end{pmatrix}.$$

- The linear model is the most fundamental of all serious statistical models underpinning:
 - ANOVA: y_i is continuous, x_{ij} 's are *all* categorical
 - REGRESSION: y_i is continuous, x_{ij} 's are continuous
 - ANCOVA: y_i is continuous, x_{ij} 's are continuous for some j and categorical for others.

Conjugate Bayesian Linear Regression: The M&m formula

- A conjugate Bayesian linear model is given by:

$$y_i | \mu_i, \sigma^2, X \stackrel{\text{ind}}{\sim} N(\mu_i, \sigma^2); \quad i = 1, 2, \dots, n;$$

$$\mu_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} = x_i^\top \beta; \quad \beta = (\beta_0, \beta_1, \dots, \beta_p)^\top;$$

$$\beta | \sigma^2 \sim N(M_0 m_0, \sigma^2 M_0); \quad \sigma^2 \sim IG(a_0, b_0).$$

- Unknown parameters include the regression parameters and the variance, i.e. $\theta = \{\beta, \sigma^2\}$.
- We assume X is observed without error and all inference is conditional on X .
- The above model is often written in terms of the posterior density $p(\theta | y) \propto p(\theta, y)$:

$$IG(\sigma^2 | a_0, b_0) \times N(\beta | M_0 m_0, \sigma^2 M_0) \times \prod_{i=1}^n N(y_i | x_i^\top \beta, \sigma^2).$$

Conjugate Bayesian (General) Linear Regression

- A more general conjugate Bayesian linear model:

$$y \mid \beta, \sigma^2, X \sim N(X\beta, \sigma^2 V_y)$$

$$\beta \mid \sigma^2 \sim N(M_0 m_0, \sigma^2 M_0) ;$$

$$\sigma^2 \sim IG(a_0, b_0) .$$

- V_y , m_0 , M_0 , a_0 and b_0 are assumed fixed.
- Unknown parameters include the regression parameters and the variance, i.e. $\theta = \{\beta, \sigma^2\}$.
- We assume X is observed without error and all inference is conditional on X .
- The posterior density $p(\theta \mid y) \propto p(\theta, y)$:

$$IG(\sigma^2 \mid a_0, b_0) \times N(\beta \mid M_0 m_0, \sigma^2 M_0) \times N(y \mid X\beta, \sigma^2 V_y)$$

- The model on the previous slide is a special case with $V_y = I_n$ ($n \times n$ identity matrix).

Deriving $p(\sigma^2 | y)$

$$\begin{aligned} p(\sigma^2 | y) &= \int p(\sigma^2, \beta | y) d\beta \\ &= IG(\sigma^2 | a_0, b_0) \times \int N(\beta | M_0 m_0, \sigma^2 M_0) \times N(y | X\beta, \sigma^2 V_y) d\beta \\ &\propto IG(\sigma^2 | a_0, b_0) \times \left(\frac{1}{\sigma^2} \right)^{\frac{n+p}{2}} \int \exp \left\{ -\frac{1}{2\sigma^2} (Q_1(\beta) + Q_2(\beta, y)) \right\} , \\ &\hspace{25em} (1) \end{aligned}$$

where

$$\begin{aligned} Q_1(\beta) &= (\beta - M_0 m_0)^\top M_0^{-1} (\beta - M_0 m_0) \quad \text{and} \\ Q_2(\beta, y) &= (y - X\beta)^\top V_y^{-1} (y - X\beta) . \end{aligned}$$

Deriving $p(\sigma^2 | y)$

$$\begin{aligned} Q_1(\beta) + Q_2(\beta, y) &= (\beta - M_0 m_0)^\top M_0^{-1} (\beta - M_0 m_0) + (y - X\beta)^\top V_y^{-1} (y - X\beta) \\ &= \beta^\top M_0^{-1} \beta - 2\beta^\top m_0 + m_0^\top M_0 m_0 \\ &\quad + \beta^\top X^\top V_y^{-1} X \beta - 2\beta^\top X^\top V_y^{-1} y + y^\top V_y^{-1} y \\ &= \beta^\top (M_0^{-1} + X^\top V_y^{-1} X) \beta - 2\beta^\top (m_0 + X^\top V_y^{-1} y) \\ &\quad + m_0^\top M_0 m_0 + y^\top V_y^{-1} y \\ &= \beta^\top M^{-1} \beta - 2\beta^\top m + c \end{aligned} \tag{2}$$

where M is a symmetric positive definite matrix, m is a vector and c is a scalar given by:

$$\begin{aligned} M^{-1} &= M_0^{-1} + X^\top V_y^{-1} X ; \\ m &= m_0 + X^\top V_y^{-1} y ; \\ c &= m_0^\top M_0 m_0 + y^\top V_y^{-1} y . \end{aligned}$$

Note: M , m and c do not depend upon β .

Deriving $p(\sigma^2 | y)$ (contd.)

Multivariate Completion of Square (ellipsoidal rectification)

$$\beta^\top M^{-1} \beta - 2\beta^\top m = (\beta - Mm)^\top M^{-1} (\beta - Mm) - m^\top Mm \quad (3)$$

Easy to verify by simply expanding the right hand side.

Another way (assuming you do not know the form on the right hand side) is to assume a centered quadratic form for the right hand side and find the coefficients. That is, write

$$\beta^\top M^{-1} \beta - 2\beta^\top m = (\beta - a)^\top B (\beta - a) + d ,$$

and try to find a (vector), B (matrix, assumed symmetric) and d (scalar) by comparing the coefficients with the left hand side. So, if the above is true then

$$\beta^\top M^{-1} \beta - 2\beta^\top m = \beta^\top B \beta - 2\beta^\top Ba + a^\top Ba + d .$$

Hence, we conclude:

$$B = M^{-1} ; Ba = m \implies a = B^{-1} m = Mm ; a^\top Ba + d = 0 \implies d = -a^\top Ba = -m^\top Mm .$$

Deriving $p(\sigma^2 | y)$ (contd.)

- Apply (3) to (2):

$$\begin{aligned} Q_1(\beta) + Q_2(\beta, y) &= (\beta - Mm)^\top M^{-1}(\beta - Mm) - m^\top Mm + c \\ &= (\beta - Mm)^\top M^{-1}(\beta - Mm) + c^*, \end{aligned} \quad (4)$$

where

$$\begin{aligned} M^{-1} &= M_0^{-1} + X^\top V_y^{-1} X ; \\ m &= m_0 + X^\top V_y^{-1} y ; \\ c &= m_0^\top M_0 m_0 + y^\top V_y^{-1} y ; \\ c^* &= c - m^\top Mm = m_0^\top M_0 m_0 + y^\top V_y^{-1} y - m^\top Mm . \end{aligned}$$

Deriving $p(\sigma^2 | y)$ (concludes)

- Apply (4) to (1):

$$\begin{aligned} p(\sigma^2 | y) &\propto IG(\sigma^2 | a_0, b_0) \times \left(\frac{1}{\sigma^2}\right)^{\frac{n+p}{2}} \int \exp \left\{ -\frac{1}{2\sigma^2} (Q_1(\beta) + Q_2(\beta, y)) \right\} \\ &\propto IG(\sigma^2 | a_0, b_0) \times \left(\frac{1}{\sigma^2}\right)^{\frac{n}{2}} \int \left(\frac{1}{\sigma^2}\right)^{\frac{p}{2}} \exp \left\{ -\frac{1}{2\sigma^2} (\beta - Mm)^\top M^{-1} (\beta - Mm) + c^* \right\} \\ &\propto \left(\frac{1}{\sigma^2}\right)^{a_0+1} \exp \left\{ -\frac{b_0}{\sigma^2} \right\} \times \left(\frac{1}{\sigma^2}\right)^{\frac{n}{2}} \exp \left\{ -\frac{c^*}{2\sigma^2} \right\} \times \int N(\beta | Mm, \sigma^2 M) \\ &\propto \left(\frac{1}{\sigma^2}\right)^{a_0+\frac{n}{2}+1} \exp \left\{ -\frac{1}{\sigma^2} \left(b_0 + \frac{c^*}{2} \right) \right\} . \end{aligned}$$

- Therefore,

$$p(\sigma^2 | y) = IG(\sigma^2 | a^*, b^*)$$

where

$$a^* = a_0 + \frac{n}{2} ;$$

$$b^* = b_0 + \frac{c^*}{2} = b_0 + \frac{1}{2} \{ m_0^\top M_0 m_0 + y^\top V_y^{-1} y - m^\top Mm \}$$

Conjugate Bayesian (General) Linear Regression

- The joint posterior density can be written as

$$p(\beta, \sigma^2 | y) \propto \underbrace{IG(\sigma^2 | a^*, b^*)}_{p(\sigma^2 | y)} \times \underbrace{N(\beta | Mm, \sigma^2 M)}_{p(\beta | \sigma^2, y)},$$

where

$$a^* = a_0 + \frac{n}{2}; \quad b^* = b_0 + \frac{1}{2} (m_0^\top M_0 m_0 + y^\top V_y^{-1} y - m^\top M m);$$

$$m = m_0 + X^\top V_y^{-1} y; \quad M^{-1} = M_0^{-1} + X^\top V_y^{-1} X.$$

Algorithm Exact posterior sampling from $p(\beta, \sigma^2 | y)$

- 1: **for** (i in $1 : N$), **do**
 - 2: Draw $\sigma_{(i)}^2 \sim IG(a^*, b^*)$
 - 3: Draw $\beta_{(i)} \sim N(Mm, \sigma_{(i)}^2 M)$
 - 4: **end for**
 - 5: **return** $(\sigma_{(i)}^2, \beta_{(i)})_{i=1,2,\dots,N}$
-

Bayesian predictions from linear regression

- Let $\tilde{y}$ denote an $m \times 1$ vector of outcomes we seek to predict based upon predictors $\tilde{X}$.
- We wish to draw samples from:

$$p(\tilde{y} | y) = \int p(\tilde{y} | \theta, y) p(\theta | y) d\theta$$

- Or, equivalently, draw from joint $p(\theta, \tilde{y} | y)$, using composition and save samples of $\tilde{y}$:

Algorithm Exact posterior predictive sampling from $p(\tilde{y} | y)$

```
1: for ( $i$  in  $1 : N$ ), do  
2:   Draw  $\theta_{(i)} \sim p(\theta | y)$   
3:   Draw  $\tilde{y}_{(i)} \sim p(\tilde{y} | \theta_{(i)}, y)$   
4: end for  
5: return  $(\tilde{y}_{(i)})_{i=1,2,\dots,N}$ 
```

Bayesian predictions from linear regression (contd.)

- For legitimate probabilistic predictions (forecasting), the conditional distribution $p(\tilde{y} | \theta, y)$ must be well-defined.
- For example, consider the case with $V_y = I_n$. Specify the linear model:

$$\begin{bmatrix} y \\ \tilde{y} \end{bmatrix} = \begin{bmatrix} X \\ \tilde{X} \end{bmatrix} \beta + \begin{bmatrix} \epsilon \\ \tilde{\epsilon} \end{bmatrix} ; \quad \begin{bmatrix} \epsilon \\ \tilde{\epsilon} \end{bmatrix} \sim N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \sigma^2 \begin{bmatrix} I_n & O \\ O & I_m \end{bmatrix} \right) .$$

- Easy to derive the conditional density:

$$p(\tilde{y} | \theta, y) = p(\tilde{y} | \theta) = N(\tilde{y} | \tilde{X}\beta, \sigma^2 I_m)$$

- Posterior predictive density:

$$p(\tilde{y} | y) = \int N(\tilde{y} | \tilde{X}\beta, \sigma^2 I_m) p(\beta, \sigma^2 | y) d\beta d\sigma^2 .$$

Bayesian predictions from linear regression (algorithm)

Algorithm Exact posterior predictive sampling from $p(\tilde{y} | y)$

- 1: **for** (i in $1 : N$), **do**
 - 2: Draw $\sigma_{(i)}^2 \sim IG(a^*, b^*)$
 - 3: Draw $\beta_{(i)} \sim N(Mm, \sigma_{(i)}^2 M)$
 - 4: Draw $\tilde{y}_{(i)} \sim N(\tilde{X}\beta_{(i)}, \sigma_{(i)}^2 I_m)$
 - 5: **end for**
 - 6: **return** $(\tilde{y}_{(i)})_{i=1,2,\dots,N}$
-

Bayesian predictions from general linear regression

- For example, consider the case with general V_y . Specify:

$$\begin{bmatrix} y \\ \tilde{y} \end{bmatrix} = \begin{bmatrix} X \\ \tilde{X} \end{bmatrix} \beta + \begin{bmatrix} \epsilon \\ \tilde{\epsilon} \end{bmatrix} ; \quad \begin{bmatrix} \epsilon \\ \tilde{\epsilon} \end{bmatrix} \sim N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \sigma^2 \begin{bmatrix} V_y & V_{y\tilde{y}} \\ V_{y\tilde{y}}^\top & V_{\tilde{y}} \end{bmatrix} \right) .$$

- Derive the conditional density $p(\tilde{y} | \theta, y) = N(\tilde{y} | \mu_{\tilde{y}|y}, \sigma^2 V_{\tilde{y}|y})$:

$$\mu_{\tilde{y}|y} = \tilde{X}\beta + V_{y\tilde{y}}^\top V_y^{-1}(y - X\beta) ; \quad V_{\tilde{y}|y} = V_{\tilde{y}} - V_{y\tilde{y}}^\top V_y^{-1} V_{y\tilde{y}} .$$

- Posterior predictive density:

$$p(\tilde{y} | y) = \int N(\tilde{y} | \mu_{\tilde{y}|y}, \sigma^2 V_{\tilde{y}|y}) p(\beta, \sigma^2 | y) d\beta d\sigma^2 .$$

Bayesian predictions from general linear regression (algorithm)

Algorithm Exact posterior predictive sampling from $p(\tilde{y} | y)$

- 1: **for** (i in $1 : N$), **do**
 - 2: Draw $\sigma_{(i)}^2 \sim IG(a^*, b^*)$
 - 3: Draw $\beta_{(i)} \sim N(Mm, \sigma_{(i)}^2 M)$
 - 4: Compute $\mu_{\tilde{y}|y}$ using $\beta_{(i)}$ and draw $\tilde{y}_{(i)} \sim N(\mu_{\tilde{y}|y}, \sigma_{(i)}^2 V_{\tilde{y}})$
 - 5: **end for**
 - 6: **return** $(\tilde{y}_{(i)})_{i=1,2,\dots,N}$
-

Bayesian Regression Analysis for BIG DATA

Bayesian updating with independent data

- Sequential update for $\theta = \{\beta, \sigma^2\}$: Update posterior for $t = 1, 2, \dots, T$

$$\begin{aligned} p(\theta \mid y_1, \dots, y_t) &\propto p(\theta, y_1, \dots, y_{t-1}, y_t) \\ &\propto \underbrace{p(\theta, y_1, \dots, y_{t-1})}_{p(y_1, \dots, y_{t-1}) \times p(\theta \mid y_1, \dots, y_{t-1})} \times \underbrace{p(y_t \mid \theta, y_1, \dots, y_{t-1})}_{p(y_t \mid \theta)} \\ &\propto p(\theta \mid y_1, \dots, y_{t-1}) \times p(y_t \mid \theta) . \end{aligned}$$

- Note the use of independent data given θ :

$$p(y_t \mid \theta, y_1, \dots, y_{t-1}) = p(y_t \mid \theta)$$

- Posterior at time $t - 1$ becomes the prior for the next t .
- Divide & Conquer by loading manageable blocks of data for each t .

Bayesian regression using Divide and Conquer

- Partition data as $\{y_t, X_t\}$, $t = 1, 2, \dots, T$, y_t is $n_t \times 1$, X_t is $n_t \times p$ and $n = \sum_{t=1}^T n_t$.
- Suppose we assume block independence:

$$y_t | \beta, \sigma^2 \stackrel{\text{ind}}{\sim} N(X_t \beta, \sigma^2 V_t);$$
$$\beta | \sigma^2 \sim N(M_0 m_0, \sigma^2 M_0); \quad \sigma^2 \sim IG(a_0, b_0) .$$

- Let $\theta = \{\beta, \sigma^2\}$. The posterior is $p(\theta | y) \propto p(\theta, y)$:

$$IG(\sigma^2 | a_0, b_0) \times N(\beta | M_0 m_0, \sigma^2 M_0) \times \prod_{t=1}^T N(y_t | X_t \beta, \sigma^2 V_t) .$$

- Note: If each data point is independent, then $V_t = I_{n_t}$. Otherwise, we can try to design data collection in a manner that data are independent across blocks.

Bayesian regression using Divide and Conquer

- Write the posterior distribution at time $t - 1$ as

$$p(\theta | y_1, \dots, y_{t-1}) = IG(\sigma^2 | a_{t-1}, b_{t-1}) \times N(\beta | M_{t-1}m_{t-1}, \sigma^2 M_{t-1})$$

- Posterior at time t updates as

$$p(\theta | y_1, \dots, y_t) = IG(\sigma^2 | a_t, b_t) \times N(\beta | M_t m_t, \sigma^2 M_t)$$

where a_t , b_t , m_t and M_t are updated as follows:

$$m_t = m_{t-1} + X_t^\top V_t^{-1} y_t; \quad M_t^{-1} = M_{t-1}^{-1} + X_t^\top V_t^{-1} X_t$$

$$a_t = a_{t-1} + \frac{n_t}{2}; \quad b_t = b_{t-1} + \frac{q_t}{2}, \quad \text{where}$$

$$q_t = y_t^\top y_t + m_{t-1}^\top M_{t-1} m_{t-1} - m_t^\top M_t m_t.$$

Bayesian analysis for BIG DATA (algorithm)

Algorithm Exact posterior sampling for Divide & Conquer

```
1: Set  $m \leftarrow m_0$ ,  $M^{-1} \leftarrow M_0^{-1}$ ,  $a \leftarrow a_0$ ,  $b \leftarrow b_0$ 
2:  $M \leftarrow \text{solve}(M^{-1}, I_p)$ 
3: for ( $t$  in  $1 : T$ ), do
4:    $q \leftarrow \|y_t\|^2 + m^\top M m$ 
5:    $m \leftarrow m + X_t^\top V_t^{-1} y_t$  and  $M^{-1} \leftarrow M^{-1} + X_t^\top V_t^{-1} X_t$ 
6:    $q \leftarrow q - m^\top M m$ 
7:    $b \leftarrow b + q/2$  and  $a \leftarrow a + n_t/2$ 
8: end for
9:
10: for ( $i$  in  $1 : N$ ), do
11:   Draw  $\sigma_{(i)}^2 \sim IG(a, b)$ 
12:   Draw  $\beta_{(i)} \sim N(Mm, \sigma_{(i)}^2 M)$ 
13: end for
14: return  $\left\{ \beta_{(i)}, \sigma_{(i)}^2 \right\}_{i=1, \dots, N}$ 
```

Bayesian analysis for BIG DATA (algorithm using linear algebra)

Algorithm Exact posterior sampling for Divide & Conquer

```
1: Set  $m \leftarrow m_0$ ,  $M^{-1} \leftarrow M_0^{-1}$ ,  $a \leftarrow a_0$ ,  $b \leftarrow b_0$ 
2:  $L^{-1} \leftarrow \text{cholesky}(M^{-1})$  and  $L = \text{trsolve}(L^{-1}, I_p)$ 
3: for ( $t$  in  $1 : T$ ), do
4:    $q \leftarrow \|y_t\|^2 + \|Lm\|^2$ 
5:    $m \leftarrow m + X_t^\top V_t^{-1} y_t$  and  $M^{-1} \leftarrow M^{-1} + X_t^\top V_t^{-1} X_t$ 
6:    $L^{-1} \leftarrow \text{cholesky}(M^{-1})$  and  $L = \text{trsolve}(L^{-1}, I_p)$ 
7:    $q \leftarrow q - \|Lm\|^2$ 
8:    $b \leftarrow b + q/2$  and  $a \leftarrow a + n_t/2$ 
9: end for
10:
11: for ( $i$  in  $1 : N$ ), do
12:   Draw  $\sigma_{(i)}^2 \sim IG(a, b)$  and  $z \sim N(0, I_p)$ 
13:    $\beta_{(i)} \leftarrow L^\top Lm + \sigma_{(i)} L^\top z$ 
14: end for
15: return  $\{\beta_{(i)}, \sigma_{(i)}^2\}_{i=1, \dots, N}$ 
```

Federated Learning: Decentralized Divide & Conquer

- Federated Learning paradigm: A central server communicates with T clients (local servers). The raw data (y_t, X_t) cannot leave client t .
- The central server holds prior parameters: M_0^{-1} , m_0 , a_0 , and b_0 .
- To achieve the exact same global posterior without centralizing the data, we exploit the additive nature of Bayesian linear regression using conjugate priors.
- Instead of passing a sequentially updated prior from one block to the next, the server broadcasts the global prior, clients compute local summary statistics in parallel, and the server aggregates them in one shot.

Federated Learning: Local Client Computations

- The client computes three localized sufficient statistics.
- Precision Matrix Update: $\Delta M_t^{-1} = X_t^\top V_t^{-1} X_t$.
- Mean Vector Update: $\Delta m_t = X_t^\top V_t^{-1} y_t$.
- Sum of Squares: $\Delta Q_t = y_t^\top V_t^{-1} y_t$.
- Client t then sends only the tuple $(\Delta M_t^{-1}, \Delta m_t, \Delta Q_t, n_t)$ back to the central server.
- This preserves privacy and vastly reduces communication overhead. For a dataset with millions of rows but $p = 10$ predictors, the client only transmits a 10×10 matrix, a 10×1 vector, and two scalars.

Federated Learning: Server Aggregation

- The global precision matrix and natural mean vector are:

$$M_T^{-1} = M_0^{-1} + \sum_{t=1}^T \Delta M_t^{-1} \quad m_T = m_0 + \sum_{t=1}^T \Delta m_t .$$

- The shape parameter for the Inverse-Gamma distribution updates:

$$a_T = a_0 + \frac{1}{2} \sum_{t=1}^T n_t$$

- Sum across all T clients *telescopes* to yield b_T :

$$\sum_{t=1}^T q_t = \left(\sum_{t=1}^T \Delta Q_t \right) + m_0^\top M_0 m_0 - m_T^\top M_T m_T ; \quad b_T = b_0 + \frac{1}{2} \sum_{t=1}^T q_t .$$

- The server computes the global scale parameter in one step:

$$b_T = b_0 + \frac{1}{2} \left(\sum_{t=1}^T \Delta Q_t + m_0^\top M_0 m_0 - m_T^\top M_T m_T \right) .$$

Global Posterior Sampling

- With M_T , m_T , a_T , and b_T fully assembled at the central server, the global posterior is mathematically indistinguishable from a model that centralized all data in one place.

Algorithm Draw N from $p(\beta, \sigma^2 | y)$

```
1: for ( $i$  in  $1 : N$ ), do  
2:   Draw  $\sigma_{(i)}^2 \sim IG(a_T, b_T)$   
3:   Draw  $\beta_{(i)} \sim N(M_T m_T, \sigma_{(i)}^2 M_T)$   
4: end for  
5: return  $(\sigma_{(i)}^2, \beta_{(i)})_{i=1,2,\dots,N}$ 
```
